

THERMISTOR CIRCUIT: HOW MACHINES FEEL TEMPERATURE





**ANANYA WATCHED
PAPA BOIL MILK...
"WHAT IF A MACHINE
COULD RING AN
ALARM
BEFORE IT SPILLS?"**



MEET THERMO THE HEAT DETECTIVE!

 Thermo on Ice!

Brrr! Thermo is shivering and holds the doors CLOSED. No power can pass through. The machine stays OFF. Thermo is keeping the gate shut!

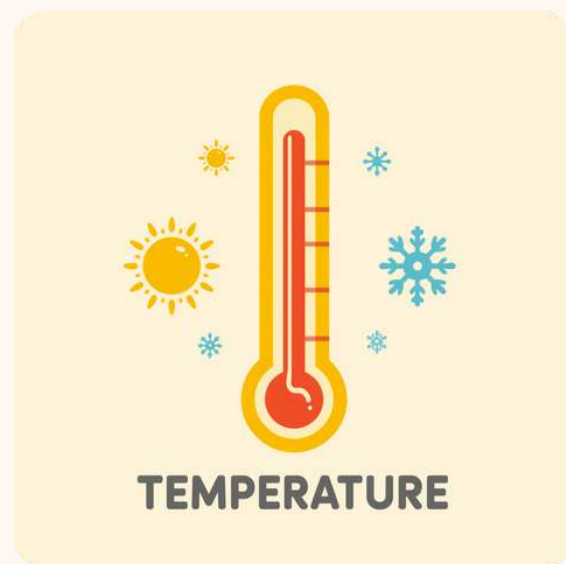


THERMO FEELS WARM!

Thermo is smiling and wearing cool sunglasses!  He throws the doors OPEN and lets power flow through. The machine turns ON. Thermo is a happy gatekeeper!

KEY WORDS TO KNOW

Let's learn some important words before we explore Thermo's world! These four words will help you understand how machines feel temperature.



Circuit

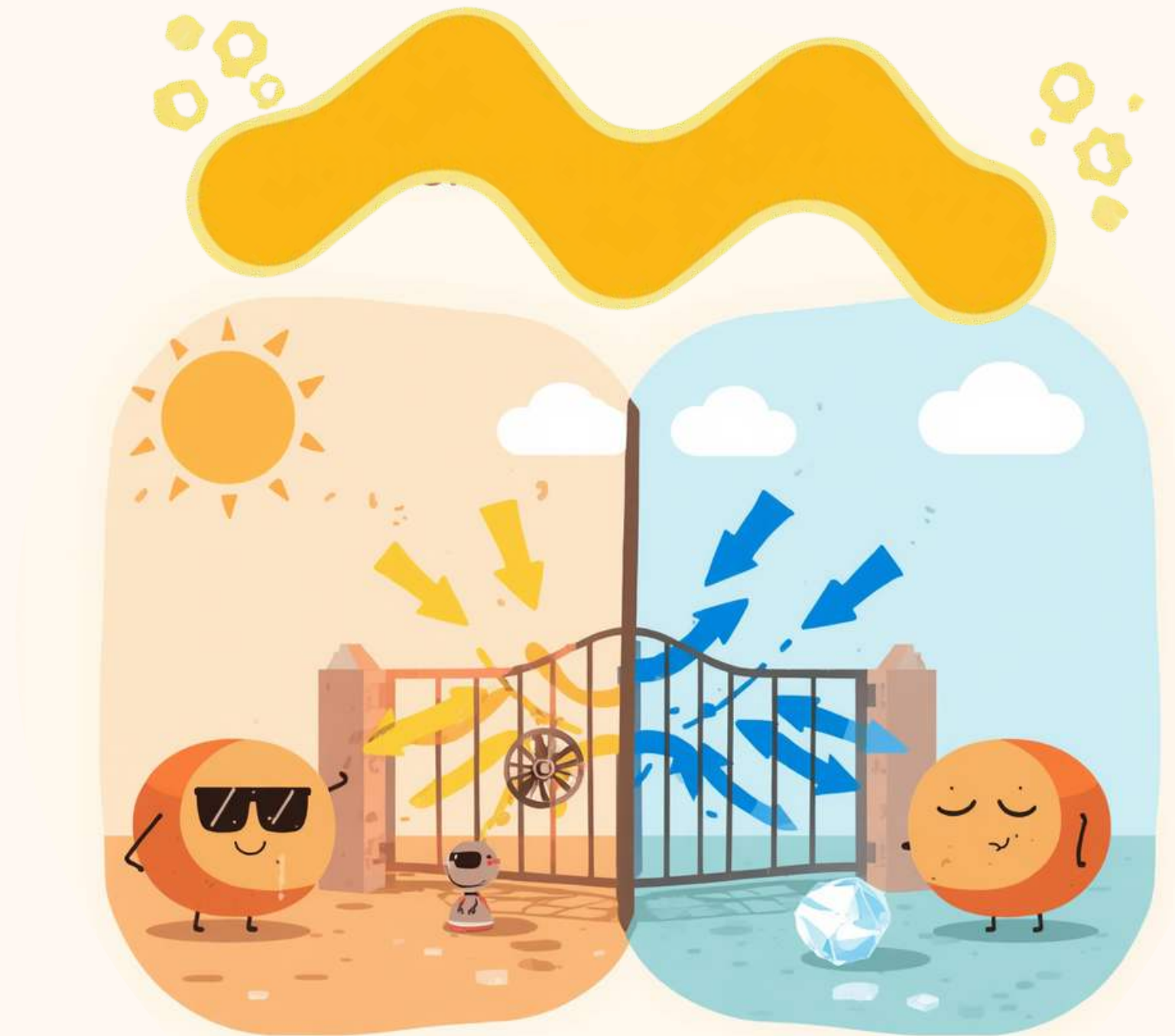
WHAT DOES A THERMISTOR DO?

Thermo watches the temperature and changes the path for electricity!

☀️ Warm = Thermo gets excited and lets power flow → machine turns ON

❄️ Cool = Thermo stays relaxed and blocks power → machine stays OFF

Just like a little gatekeeper — Thermo opens or closes the door for electricity!



HOW WE PLAY WITH THERMO!

Imagine a cool breeze blowing... now picture a hot summer sun! 🌬️☀️

Slide the scale from "Cool Breeze" to "Hot Summer Sun" and watch what happens — Thermo wakes up, the fan starts spinning, and the circuit comes alive!

Cool = Thermo sleeps, fan stays OFF.

Hot = Thermo wakes up, fan turns ON!

Try this in class: feel a cold ice cube, then a warm cup — can YOU be Thermo? 🌡️



THERMISTORS IN OUR HOMES



REFRIGERATOR

Keeps kulfi and food frozen! Thermo checks if it gets too warm and turns the cooling back on.



ELECTRIC IRON (ISTRI)

Clicks OFF automatically when it gets hot enough. Thermo protects your clothes from burning!

POWER INVERTER

Beeps and shuts off if it gets too hot during summer power cuts. Thermo keeps it safe!





MORE THERMISTOR FRIENDS AROUND YOU



SMART ACS

Slows down cooling
when the room feels
just right — especially
at night!



LAPTOP COOLING FANS

Starts spinning when your
laptop gets warm to keep it
cool and happy!

WATER HEATER / GEYSER

Turns OFF
automatically when
water is hot enough.
No more waiting!



QUIZ QUESTION 1

WHICH ONE USES THERMO TO KEEP YOU SAFE?

Look at the pictures carefully!

🔥 Option A: Clothes Iron

🕯️ Option B: Open Flame

Hint: Thermo lives inside machines that turn OFF automatically when they get too hot!

HELP RAHUL PICK THE RIGHT ANSWER!

✅ Answer: Clothes Iron! Thermo inside the iron watches the heat and clicks it OFF to keep your clothes and hands safe. An open flame has no Thermo — always stay away from fire!



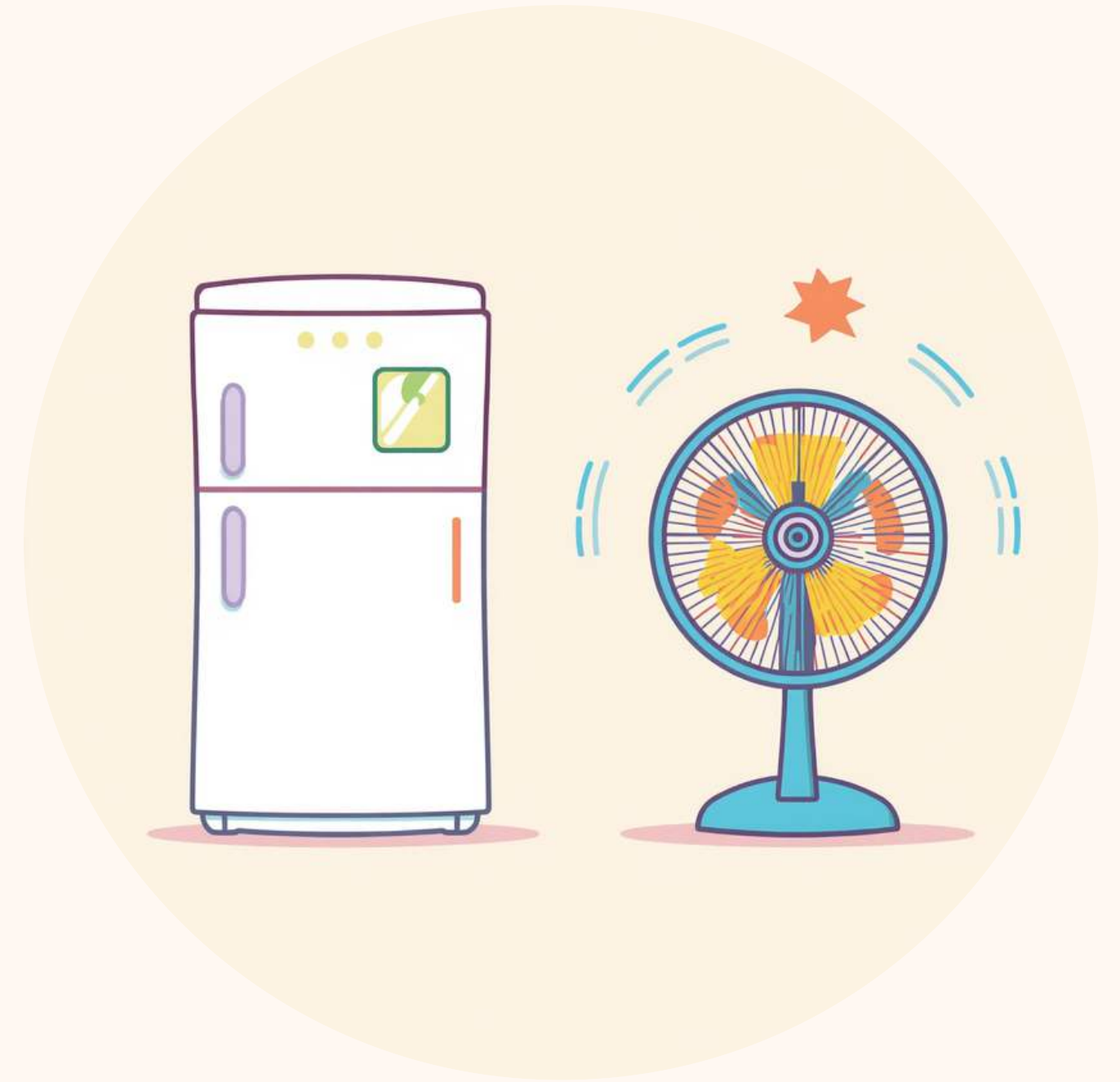
QUIZ QUESTION 2

WHICH ONE USES THERMO TO KEEP FOOD COOL?

Look at the two machines above. One of them has a thermistor (Thermo!) inside that watches the temperature and keeps your food fresh and cold. Can you guess which one?

OPTIONS: REFRIGERATOR 🧊 / ELECTRIC FAN 🌀

Hint: Thermo helps this machine stay at just the right coolness so your kulfi and food don't melt! ✅ Answer: Refrigerator



QUIZ QUESTION 3

WHICH ONE BEEPS IF IT GETS TOO HOT?

Look at the two objects on the right. One of them has Thermo inside — and it beeps loudly during hot summer power cuts to stay safe. Can you guess which one?

YOUR OPTIONS:

⚡ Power Inverter

🚗 Toy Car

✅ Answer: Power Inverter — Thermo lives inside and warns us when it overheats!



REMEMBER THE THERMO RULE!

🔥 More Heat = Thermo wakes up and lets power pass → Machine turns ON!

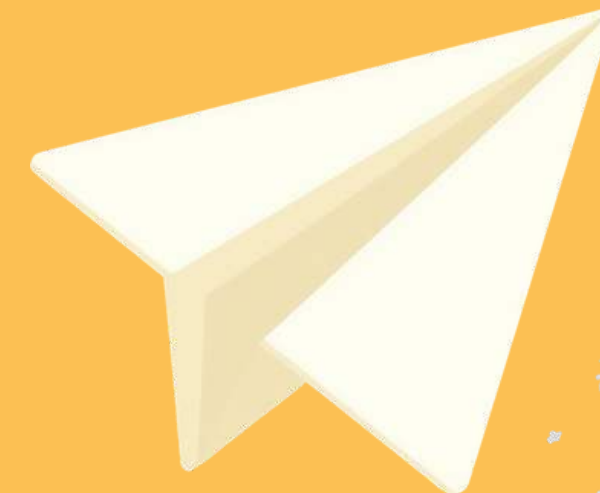
❄️ Cool = Thermo stays relaxed and blocks power → Machine stays OFF!

Thermo always watches the temperature — like a tiny, smart helper inside every machine!





YOUR TAKE-HOME MISSION!



MISSION 1: FIND THE IRON SWITCH

Ask your parents to show you the automatic switch on a clothes iron (istri). Watch how it works when it gets hot!

MISSION 2: LISTEN FOR THE CLICK!

Listen carefully for the "click" sound when the iron turns itself OFF. That is Thermo doing its job — keeping you safe!

MISSION 3: IMAGINE YOUR THERMO!

Close your eyes and imagine your own little Thermo watching the temperature inside machines at home. What would Thermo say?



THANK YOU, HEAT DETECTIVES!

Thank you for learning about Thermo the Heat Detective! Any questions or stories about machines that feel heat?

